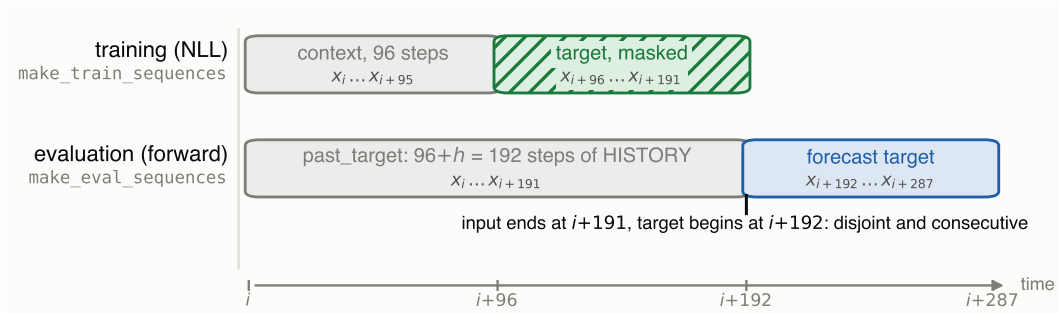



Both rows are the index arrays the two builders return for window *i*, drawn to one shared scale. At evaluation the whole input is passed as past_target with past_observed_target all ones, and no target value is anywhere in it. Training's two halves are concatenated into a single 96+*h* tensor, but _val_loss splits it at context_length and the module substitutes its learned mask_encoding at every target position, so the target is scored by the likelihood and never conditioned on.

Figure S1: **Which timesteps Moirai is given, and which it is asked to predict**, at $h$=96; $h$=192 is the same layout with $h$ substituted. Both rows are the index arrays that `make_train_sequences` and `make_eval_sequences` return for window $i$, drawn to one shared scale. At evaluation the model input is the $96{+}h$ steps *preceding* the forecast window and the target is the $h$ steps strictly after it: the two are disjoint and consecutive, so no target value reaches the model. The figure is generated from those functions' output and asserts that disjointness at draw time.

# SUPPLEMENTARY MATERIAL: WHEN REPRESENTATION CHANGE MISLEADS

**Anonymous authors**
Paper under double-blind review

Detailed robustness analyses referenced from the main paper's body and appendix. Section, table and figure numbers here carry an "S" prefix; numbers without it refer to the main paper.

## S1 ETTH2 FORECASTING DETAILS

**Dataset.** ETTh2 (Electricity Transformer Temperature, Hourly, dataset 2) contains 17,420 timesteps of 7 features (HUFL, HULL, MUFL, MULL, LUFL, LULL, OT) recorded hourly from electrical transformers. We use the standard chronological split: 12 months training, 4 months validation, 4 months test.

**Window construction, and exactly which values Moirai is given.** The evaluation windows use an "extended lookback" of $96{+}h$ steps, which could be read as though the forecast target were part of the model input. It is not, and because the training and evaluation window sets differ the arithmetic is worth writing out. Let $x_1, \dots, x_T$ be the series and $h$ the horizon. Both window sets are built by two functions in `scripts/finetune_forecasting.py`, named here so that this paragraph and the code can be checked against each other.

*Training* (`make_train_sequences`): window $i$ has context $x_i, \dots, x_{i+95}$ and target $x_{i+96}, \dots, x_{i+95+h}$. The two are concatenated into one $96{+}h$ tensor and passed to Moirai's `_val_loss` with fixed patch size 32, which is the interface's required calling convention: `_val_loss` splits the tensor at `context_length` and marks the latter part with `prediction_mask`, and the module then substitutes its learned `mask_encoding` at every masked position and zeroes the corresponding observed flags. The target values are therefore scored by the likelihood and never conditioned on.

*Evaluation* (`make_eval_sequences`): window $i$ has input $x_i, \ldots, x_{i+95+h}$ (that is, $96{+}h$ steps of *history*), and target $x_{i+96+h}, \ldots, x_{i+95+2h}$. The input's last index is $i+95+h$ and the target's first is $i+96+h$, so the two windows are disjoint and consecutive and no target value appears anywhere in the model input. Only the input is passed, as `past_target` with `past_observed_target` all ones; the point prediction is the median of 20 forecast samples with `patch_size='auto'`. "Extended lookback" thus means a longer *history*, not history-plus-target. Figure S1 draws both layouts from the index arrays those two functions actually return rather than from this description, and asserts the disjointness at draw time, so the figure cannot drift from the implementation.

The one asymmetry this does introduce is between models: Moirai is given $96{+}h$ steps of history where the fitted baseline of Appendix A12 is given 96. That direction favours the pre-trained model (more context lowers $\text{MSE}_{\text{ZS}}$, raises $V_b$ and so admits *more* cells), and Appendix S8.3 measures it rather than arguing it, by giving the baseline the same $96{+}h$ history.

All data is fed to Moirai in raw scale (Moirai's internal PackedStdScaler handles instance normalization); predictions and targets are then normalized using training-set mean and standard deviation for MSE/MAE computation, matching the standard time-series forecasting evaluation protocol.

**Horizon selection.** We evaluate $h{=}96$ and $h{=}192$, the two horizons at which Moirai-Small on ETTh2 clears the value gate: measured against the fitted lookback-96 baseline the gate is defined with (Appendix A12), $V_{\text{ridge}}{=}{+}0.400$ at $h{=}96$ and $+0.464$ at $h{=}192$ on held-out windows. Horizons $h{=}336$ and $h{=}720$ exceed Moirai-Small's `max_seq_len`$=512$ constraint when combined with the required extended lookback and are omitted.

## S2 uni2ts PackedStdScaler Patch Details

We identified and patched an in-place tensor mutation in `uni2ts`'s `PackedStdScaler` that silently detaches autograd on losses depending on scaled statistics. The issue arises when `PackedStdScaler.transform` performs an in-place `.copy_()` that breaks the gradient tape. Our patch replaces the in-place write with a differentiable `torch.where` operation. *All* Moirai fine-tuning results in this paper use the patched scaler (released with our code); unpatched `uni2ts` runs may converge differently and we have observed silent non-convergence on `uni2ts@main`. We recommend upstreaming this patch to `uni2ts`.

## S3 Moirai Median Forecast Selection

Moirai generates probabilistic forecasts as a mixture of distributions (StudentT, NormalFixedScale, NegativeBinomial, LogNormal). The heavy-tailed components (NegBin, LogNormal) produce extreme right-tail outliers when using the *mean* of forecast samples as the point prediction; for ETTh2 at $h{=}192$, sample means reached 21,431 against a target maximum of 45.5, inflating MSE from 0.30 to 8.58.

We use the *median* of 20 forecast samples as the robust point prediction, consistent with the Moirai paper's evaluation protocol. This eliminates sensitivity to the heavy-tailed mixture components while preserving the distributional forecast's central tendency.

## S4 The Pre-Registered Prospective Test, in Full

**What was registered, and when.** Predictions were written to `results/v47_prospective/preregistration.json` and committed to git at a named revision before any outcome run for those cells existed; the scoring script reads that file and the finished runs and decides nothing. Two rules were registered against each other: the **gate rule**, $\text{gate}_{\text{pre}} \geq 0.20 \Rightarrow$ at risk, which is the paper's own criterion frozen while the screen still used the unfitted trend denominator, and therefore graded on those scores rather than re-derived after seeing the outcomes; and a **dataset rule**, dataset $\in$ {ETTh1, Weather} $\Rightarrow$ degradation, a *post hoc* pattern read off the original Moirai cells and registered as a competitor so the gate is graded against something. Eight cells were then run (Weather, ETTm2, Electricity7, ETTh1 $\times$ Small/Base/Large, 3 seeds each) and scored by the pre-registered outcome definition, unchanged (Table S1).

**Result: the outcome the gate was registered to predict never occurs.** None of the eight cells degraded. The gate flagged all eight, giving **TP 0, FP 8, FN 0**: precision $0.00$, while **sensitivity and specificity are both unestimable**: sensitivity has an empty positive class in its denominator, and specificity is $0/8$ only because every prediction was positive. The competing dataset rule scores no better, for the same reason. **What "prospective" does and does not mean here:** the predictions were frozen before these cells were *trained*, so they are prospective with respect to the intervention, but the datasets and their splits already existed, so this is not a test on newly collected data. Nor is the empty positive class an artefact of the fitted screen: only one of the eight clears $0.20$ on either split under it (base/ETTm2 $h{=}192$, $+0.21$ on selection windows) and it fails clause (ii) in 0 of 3 seeds. **Nor is the $0.20$ threshold what fails**: re-scoring the same predictor against the same outcome at thresholds from $0.10$ to $0.60$, moving nothing else, gives precision $0.00$ at every one, with 8, 8, 6, 5, 3 and 2 cells flagged (Table S2); the problem is not the cutoff but that the predicted class is empty.

**What a zero-prevalence test licenses.** With no positive instance the arm is informative in one direction only. It establishes that the gate's positive predictions are wrong here, at every threshold swept, and that the error is not an artefact of the operating point. It establishes *nothing* about whether the gate would detect degradation if degradation existed: that is a claim about sensitivity, and $0/0$ is not a rate. A reader who takes this arm as a refutation of the gate is reading past the prevalence. The regime the arm cannot reach is one a positive control has to construct deliberately (§7.1, Appendix A18), and which, on these benchmarks, we could not construct.

**The gate does not provide a reliable ordering, and there is no actionable threshold.** Within Moirai ($n{=}22$ cells from six series, the only arm with enough cells to ask), the gate's continuous score correlates *negatively* with held-out $\Delta_{\text{enc}}$: Spearman $\rho{=}{-}0.293$, and neither cells nor series exclude zero, so we report the direction and not the interval (Appendix A16). The sign is the wrong one (higher measured pre-trained value goes with encoder adaptation helping *more*, not less, the opposite of what a preservation heuristic needs), but the sharper point is that the magnitude is not stable: scoring the same gate on the held-out windows instead gives $\rho{=}{-}0.518$ on the same cells. An ordering that nearly doubles when the gate changes split is not one a practitioner can set a threshold on, and no other quantity we measured orders $\Delta_{\text{enc}}$ either (Appendix S13).

Table S1: **The pre-registered prospective test.** Predictions frozen in git before any condition-B or -D run for these cells existed. "Degradation" applies the paper's own definition unchanged: gate-passing $\wedge$ $\text{forg}_{\text{B}}{>}0$ in every seed $\wedge$ $\text{forg}_{\text{D}}{<}0$ in every seed. $\text{gate}_{\text{pre}}$ is the frozen predictor's score, on the unfitted denominator; $\text{gate}_{\text{val}}$ is the fitted selection-split gate of §4.

| Cell (new) | $\text{gate}_{\text{pre}}$ | $\text{gate}_{\text{val}}$ | $\text{forg}_{\text{B}}$ | B$-$D held-out | Degradation? | Predicted |
|---|---|---|---|---|---|---|
| base Weather ($h{=}96$) | $+0.41$ | $-0.82$ | $-11.7$ (0/3) | $+9.4{\pm}2.0$ | no | gate $\times$, data $\times$ |
| base Weather ($h{=}192$) | $+0.59$ | $-0.63$ | $-18.1$ (0/3) | $+12.5{\pm}8.7$ | no | gate $\times$, data $\times$ |
| base ETTm2 ($h{=}96$) | $+0.30$ | $-0.26$ | $-13.1$ (0/3) | $+7.0{\pm}2.4$ | no | gate $\times$, data ✓ |
| base ETTm2 ($h{=}192$) | $+0.26$ | $+0.21$ | $-8.4$ (0/3) | $+1.3{\pm}3.4$ | no | gate $\times$, data ✓ |
| small Electricity7 ($h{=}96$) | $+0.71$ | $-0.65$ | $-16.0$ (0/3) | $-3.2{\pm}1.5$ | no | gate $\times$, data ✓ |
| small Electricity7 ($h{=}192$) | $+0.76$ | $-0.45$ | $-18.6$ (0/3) | $-2.6{\pm}2.1$ | no | gate $\times$, data ✓ |
| large ETTh1 ($h{=}96$) | $+0.35$ | $-0.25$ | $+14.1$ (3/3) | $+28.2{\pm}4.9$ | no | gate $\times$, data $\times$ |
| large Weather ($h{=}96$) | $+0.47$ | $-0.64$ | $+7.9$ (3/3) | $+12.0{\pm}7.5$ | no | gate $\times$, data $\times$ |

Table S2: Gate threshold sensitivity on the eight pre-registered cells (§4). The predictor and the outcome definition are both as pre-registered; only the threshold moves. Bold is the operating point used in the paper.

| Threshold | Flagged | TP | FP | FN | Precision | Recall |
|---|---|---|---|---|---|---|
| 0.10 | 8 | 0 | 8 | 0 | 0.00 | $nan$ |
| 0.20 | 8 | 0 | 8 | 0 | 0.00 | $nan$ |
| 0.30 | 6 | 0 | 6 | 0 | 0.00 | $nan$ |
| 0.40 | 5 | 0 | 5 | 0 | 0.00 | $nan$ |
| 0.50 | 3 | 0 | 3 | 0 | 0.00 | $nan$ |
| 0.60 | 2 | 0 | 2 | 0 | 0.00 | $nan$ |

## S5 BATCH 3: THE FITTED GATE, REGISTERED BEFORE ITS OWN OUTCOMES

**Why a second batch, and what was registered.** The arm above grades a predictor we have since corrected, and it fired on every cell it saw, so it cannot say what happens when the gate *declines* to fire. A second batch was therefore registered in `results/v57_prospective3/preregistration_v3.json`, committed on 2026-09-22 at revision `e5f1d626`, **before any condition-B or condition-D run of any cell in it existed**: **ten** cells: five Moirai (Base/Electricity7 at $h\in\{96, 192\}$, Large/ETTh2 $h=192$, Large/ETTm2 at $h\in\{96, 192\}$) and five TimesFM-2.5 at $h=48$, at 3 seeds in each of conditions B and D, with the *corrected* ridge gate as the predictor and the paired-interval rule of §3 as the registered outcome. The two batches are reported side by side and **never pooled**: they grade different predictors under different outcome rules, and a prospective test may not move its own comparison set, so the BH adjustment runs within this batch alone.

**Result: no cell degraded, and this time the gate did not flag everything** (Table S3). Across the **9** scored cells (54 outcome runs), **0** degrade under the registered interval rule and **0** under the published unanimity definition, the two agreeing as they do on the retrospective grid. The gate fired on **2** of the 9, giving **TP 0, FP 2, FN 0, TN 7**: precision $0.00$ again, sensitivity still unestimable against an empty positive class, but **specificity** $0.78$**, which batch 1 could not estimate at all**. That number is the arm's one positive contribution and it is a measure of *restraint*, not of discrimination: at zero prevalence a screen can be credited for the cells it correctly leaves alone, and nothing can be said about the cells it would have caught. The competing dataset rule flags two different cells (TimesFM-2.5 on ETTh1 and on Weather) and scores the same matrix.

**The gate's two prospective positives are both wrong, and one of them is wrong decisively.** Large/ETTh2 $h=192$ carries a gate of $+0.92$, within $0.008$ of Moirai-Small/ETTh2 $h=192$ ($+0.932$), the strongest capability the screen admits anywhere in this paper and the very cell the positive control of §7.1 had to engineer a conflicting task against. On that cell full fine-tuning does not damage the pre-trained capability: it *improves* held-out loss by $33.2\%$ against zero-shot, with a $95\%$ interval of $[-47.2, -19.2]$ that excludes zero. The gate's most confident prospective prediction is therefore not merely unconfirmed but contradicted, in the direction opposite to the one it predicts.

**What this arm cannot support, stated in its own terms.** Both the B$-$ZS and D$-$ZS intervals exclude zero on only **2** of the 9 cells, and the primary matrix counts an inconclusive cell as not degrading, as registered, so most of the specificity above rests on cells where the data are consistent with a small effect of either sign. Restricted to the 2 decisive cells the matrix is TP 0, FP 1, FN 0, TN 1, which is not a rate anyone should quote. The secondary registered outcome, the three-way call on $\Delta_{\text{enc}}$ after BH adjustment within the batch, is **inconclusive on all 9**: zero freeze-decisive and zero adapt-decisive cells, so the batch identifies no new cell on which freezing or adapting can be called, and contradicts no call made on the published grid. A batch of nine cells at 3 seeds is a test of the screen's false-positive behaviour and of nothing else.

**One registered cell was not run, and the shortfall is disclosed rather than absorbed.** Nine of the ten registered cells were scored; the tenth, Moirai-Large/ETTm2 $h=96$, was deferred on cost grounds when the submission window would not hold its measured $\sim$16.5 h of fine-tuning, and the decision is recorded with its timestamp, its reasons and the record count at the moment it was taken in `results/v57_prospective3/deferred_cell_2026-09-24.json`, committed before the deferral took effect. Two facts make it checkable rather than a matter of trust. *No outcome for that cell existed when the decision was made* (0 condition-B and 0 condition-D records), so the choice cannot have been informed by its result; and *the cell is gate-failing* ($-0.278$), so the only matrix entry it could have supplied is a true negative, which would move specificity from $0.78$ to $0.80$. Its absence can therefore only weaken the one estimate this arm exists to produce; it cannot manufacture a favourable one. The registration is unchanged and still lists all ten cells, so the shortfall is visible by diffing it against the scored cells.

**Two disclosures about the predictor.** The five TimesFM cells' registered gate was computed on the seed-42 selection-window draw, while the multi-seed recomputation in `gate_all_cells.timesfm_gates` averages the draws of all three run seeds; the frozen value is the predictor and the recomputation is a sensitivity on it, not a replacement. Those five cells are also the only registered gates the paper's own cache does not reproduce to $10^{-9}$, and by design: their numerator was measured by loading the checkpoint, and the one other route to it agrees only to

$\sim 6\times10^{-7}$ relative, which is real confirmation but would trip a bit-equality check. The five Moirai gates are recomputed by the paper's estimator in a second pass and **all five reproduce the registered value exactly**.

Table S3: **Batch 3 of the pre-registered prospective test: the fitted gate, graded on cells whose outcomes did not exist when it was frozen.** $\text{gate}_{\text{val}}$ is the corrected selection-split ridge gate of §4, frozen in the registration; "pred." is the registered gate rule at the 0.20 operating point. $\Delta_{\text{FT}}$ and $\Delta_{\text{frozen}}$ are B−ZS and D−ZS in percent of zero-shot loss, with 95% intervals propagating the zero-shot reference's own error; "Degr.?" applies the registered interval rule. The $\Delta_{\text{enc}}$ call is BH-adjusted within this batch only. A tenth registered cell (Moirai-Large/ETTm2 $h$=96, gate −0.278) was deferred for the submission deadline and is disclosed in Appendix S5.

| Cell (batch 3) | $\text{gate}_{\text{val}}$ | pred. | $n$ | $\Delta_{\text{FT}}$ | $\Delta_{\text{frozen}}$ | Degr.? | $\Delta_{\text{enc}}$ call |
|---|---|---|---|---|---|---|---|
| large ETTh2 ($h$=192) | +0.92 | risk | 3 | −33.2 [-47.2, -19.2] | −12.9 [-25.2, -0.7] | no | inconclusive |
| large ETTm2 ($h$=192) | +0.24 | risk | 3 | −10.3 [-37.5, +16.8] | −4.9 [-22.1, +12.4] | no | inconclusive |
| TimesFM-2.5 / ETTh1 ($h$=48) | −0.05 | – | 3 | −2.6 [-5.3, +0.1] | +0.1 [-2.6, +2.8] | no | inconclusive |
| TimesFM-2.5 / ETTh2 ($h$=48) | −0.08 | – | 3 | +99.2 [-345.3, +543.7] | −6.4 [-9.1, -3.8] | no | inconclusive |
| TimesFM-2.5 / ETTm2 ($h$=48) | −0.01 | – | 3 | −29.4 [-62.1, +3.3] | −16.0 [-20.3, -11.7] | no | inconclusive |
| TimesFM-2.5 / Weather ($h$=48) | −0.08 | – | 3 | −9.1 [-15.5, -2.7] | −7.6 [-8.7, -6.4] | no | inconclusive |
| TimesFM-2.5 / Electricity ($h$=48) | +0.04 | – | 3 | +12.4 [+6.2, +18.7] | −1.3 [-3.5, +0.8] | no | inconclusive |
| base Electricity7 ($h$=96) | −0.59 | – | 3 | −3.7 [-17.2, +9.9] | −18.8 [-19.8, -17.8] | no | inconclusive |
| base Electricity7 ($h$=192) | −0.51 | – | 3 | −3.8 [-31.1, +23.5] | −19.2 [-19.9, -18.6] | no | inconclusive |

## S6 LoRA Rank and Hyperparameter Sensitivity

Table S4 shows LoRA on **Moirai-Small** is robust to rank ($r \in \{4, 8, 16, 32\}$): all ranks achieve CKA≥0.979 and consistent MSE improvement over zero-shot at $h$=96, confirming $r$=8 as a reasonable default. On **Moirai-Large** (seed 42, $h$=96, $n$=500), rank escalation alone does *not* rescue the failure mode: forgetting stays +19 to +30% across $r \in \{8, 16, 32, 64\}$. Table S5 extends the sweep to $\alpha$, target-module scope, and LR (all at $r$=8, seed 42): varying $\alpha \in \{8, 16, 32\}$ and module scope provides no rescue, but reducing LR to $1\times10^{-5}$ does (forg.= $-8.5 \pm 0.7\%$, CKA=0.989±0.003, 5 seeds): the Small-recipe transfers to Large only at a 10× lower learning rate.

Table S4: LoRA rank sensitivity on ETTh2 (condition E, 500 training samples). Moirai-Small: all ranks achieve near-identical MSE and CKA≥0.979. Moirai-Large: all ranks forget +19 to +30%, with no rank recovering Moirai-Small's negative forgetting. Mean ± std across 3 seeds (5 for $r$=8 Small); Moirai-Large single-seed ($s$=42) for $r \in \{16, 32, 64\}$ (std not reported; point estimates only), 3 seeds for $r$=8.

| Backbone | Rank | $h$ | MSE | Forg.% | CKA | Seeds |
|---|---|---|---|---|---|---|
| Small | 4 | 96 | .116±.001 | −9.5±0.9 | .992±.001 | 3 |
| Small | 4 | 192 | .146±.003 | −8.8±1.9 | .995±.002 | 2 |
| Small | 8 | 96 | .116±.001 | −9.1±1.1 | .979±.006 | 5 |
| Small | 8 | 192 | .157±.002 | −1.8±3.1 | .986±.005 | 5 |
| Small | 16 | 96 | .117±.001 | −10.2±1.2 | .987±.002 | 3 |
| Small | 16 | 192 | .155±.002 | −1.7±1.2 | .988±.001 | 3 |
| Small | 32 | 96 | .119±.001 | −8.3±0.9 | .986±.001 | 3 |
| Small | 32 | 192 | .159±.003 | +1.1±1.4 | .989±.002 | 3 |
| Large | 8 | 96 | .159±.004 | +25.7±3.2 | .973±.012 | 3 |
| Large | 16 | 96 | .154 | +22.5 | .955 | 1 |
| Large | 32 | 96 | .150 | +19.0 | .986 | 1 |
| Large | 64 | 96 | .165 | +29.9 | .984 | 1 |

## S7 Chronos-T5-Small on M4-Monthly

This appendix reports Chronos-T5-Small (T5 seq2seq, tokenised cross-entropy) on M4-Monthly, at 200 series, lookback 96 and $h$=18: conditions B and D at three seeds each. The arm was

Table S5: LoRA-Large HP ablation (ETTh2, $r$=8, $n$=500, $h$=96). Baseline (default LR=1e-4, $\alpha$=16, modules = q,v,out) forg.= $+23.7\%$, seed 42. Only LR=1e-5 rescues LoRA-Large (5 seeds); all other axes fail.

| Axis | Setting | Val MSE | Forg.% | CKA | Seeds |
|---|---|---|---|---|---|
| LR | 1e-4 (default) | .157 | +23.7 | .980 | 1 |
| | 5e-5 | .146 | +15.3 | .982 | 1 |
| | **1e-5** | **.115**$\pm$**.001** | $-$**8.5**$\pm$0.7 | **.989**$\pm$**.003** | **5** |
| $\alpha$ | 8 | .155 | +21.9 | .975 | 1 |
| | 16 (default) | .157 | +23.7 | .980 | 1 |
| | 32 | .158 | +24.3 | .984 | 1 |
| Modules | q,v,out (default) | .157 | +23.7 | .980 | 1 |
| | q,k,v | .153 | +20.4 | .987 | 1 |
| | q,k,v,out | .151 | +19.0 | .988 | 1 |

registered (`results/chronos_m4/preregistration.json`) and committed before the gate was computed, so the estimator could not be chosen after seeing whether it cleared the threshold.

**Data and splits.** The cell takes the first 200 series in file order with at least 228 observations of history and 18 test values; there is no RNG in the selection, so the set is a property of the two released CSVs, whose hashes are in `results/data_manifest.json`. Each series is cut three ways: sliding training windows over all but the last 114 observations (57,327 usable, 27 dropped for a constant context), one selection window per series, and one held-out window per series whose target is the official M4-Monthly test values. The held-out *context* does contain the 18 values used as the selection target, because M4 defines its test set as the 18 steps following the history; no held-out target is ever a training or selection target. Early stopping is on the selection windows, and the best epoch is required to be at least 1, so no run can select the pre-trained checkpoint.

Table S6: **The value gate for Chronos/M4-Monthly.** Ten denominators against one numerator (median of 20 Chronos samples, per-window $z$-scored), on both splits. A denominator worse than `constant`, the training-mean predictor, cannot support a claim that the checkpoint has something worth preserving: it inflates $V_b$ in the direction that flatters the checkpoint, which is the effect that moves 17 of 21 Moirai cells between the unfitted and fitted screens (Appendix A12).

| | Selection split (primary) | | Held-out split | | |
|---|---|---|---|---|---|
| Denominator | MSE | $V_b$ | MSE | $V_b$ | Admissible |
| trend (unfitted) | 1.801 | +0.448 | 2.187 | +0.590 | — |
| fitted ridge (**primary**) | 1.095 | +0.092 | 1.132 | +0.209 | ✓ |
| constant (floor) | 1.739 | +0.428 | 2.514 | +0.644 | floor |
| persistence | 1.087 | +0.085 | 1.364 | +0.343 | ✓ |
| seasonal naive | 1.256 | +0.208 | 1.380 | +0.351 | ✓ |
| AR | 1.095 | +0.092 | 1.132 | +0.209 | ✓ |
| DLinear | 1.095 | +0.092 | 1.132 | +0.209 | ✓ |
| ridge, tuned | 1.095 | +0.092 | 1.132 | +0.209 | ✓ |
| MLP | 1.054 | +0.057 | 0.855 | $-$0.048 | ✓ |
| GBM | 1.003 | +0.008 | 0.983 | +0.089 | ✓ |

The admissibility column is the *selection* split, the split every admission decision is made on. trend (unfitted) is admissible on the held-out split and not on the selection split, the constant floor being 2.514 there against 1.739 here.

AR, DLinear and the tuned ridge agree with the primary denominator to eight significant figures, which is not a coincidence and not an error: M4-Monthly is univariate, so all four reduce to the same least-squares map $\mathbb{R}^{96} \to \mathbb{R}^{18}$.

**The gate.** Table S6 gives the ladder. On the selection split (the split every admission decision in this paper is made on), the primary fitted ridge scores $V_{\text{ridge}}$=+0.092, which *fails* the 0.20 operating point. The unfitted trend denominator reads +0.448 on those same windows and is **inadmissible**: at

an MSE of 1.801 it is worse than the constant predictor's 1.739, so what it reports is an advantage over a baseline a training mean beats. The best admissible rung is the seasonal naive at $+0.208$, the only admissible denominator this checkpoint clears; the weakest advantage on the ladder is GBM's $+0.008$. On the held-out split the fitted gate reads $+0.209$ and would pass, but the MLP rung reads $-0.048$, so on neither split does the cell survive every admissible denominator. The registration fixed the reading of this outcome before the gate was computed: **no non-Moirai cell in reach clears its own screen**, a finding about this benchmark-and-backbone pairing, not a gap in the design. The screen's disqualifier reading (§4) is unaffected; what is gone is any prospect of a non-Moirai cell with something to preserve.

Table S7: **Conditions B and D on Chronos/M4-Monthly**, $n$=500 training windows, lr $10^{-5}$, batch 32, 20 epochs with patience 5, early stopping on the selection windows and best epoch $\geq$1. Forgetting is % change in held-out MSE against the same seed's zero-shot measurement (*negative is better than zero-shot*); $\Delta_{\text{enc}}$ is the paired difference. CKA is the linear CKA of the mean-pooled final encoder output on the 200 selection contexts, and the $\ell_2$ column is the encoder state dict's distance from pre-trained. Under condition D both are fixed *by construction* and not by measurement: on Chronos, freezing the encoder freezes the shared token embedding with it.

| | Held-out forgetting | | | CKA / encoder $\ell_2$ | | |
|---|---|---|---|---|---|---|
| Seed | B (full FT) | D (frozen enc.) | $\Delta_{\text{enc}}$ | B | D | Best epoch B / D |
| 42 | $-6.52\%$ | $-5.99\%$ | $-0.53$ | 0.9967 / 1.4 | 1.0000 / 0.0 | 7 / 8 |
| 43 | $-2.31\%$ | $+1.23\%$ | $-3.54$ | 0.9960 / 1.4 | 1.0000 / 0.0 | 7 / 10 |
| 44 | $+2.64\%$ | $-2.05\%$ | $+4.69$ | 0.9941 / 1.8 | 1.0000 / 0.0 | 9 / 3 |
| mean | $-2.06\%$ | $-2.27\%$ | $+0.21 \pm 2.40$ | 95% CI $-10.14$ to $+10.56$, inconclusive | | |

**Conditions B and D.** Table S7 gives the six runs. Both arms improve on zero-shot on average (full fine-tuning by $-2.1\%$ and decoder-only adaptation by $-2.3\%$), and the paired contrast is $\Delta_{\text{enc}}$=$+0.21$ with a 95% interval of $[-10.14, +10.56]$, which is **inconclusive** under the three-way rule of §5.4. At three seeds the minimum detectable effect here is 10.35 pp, so that interval is a statement about this cell's power and not about its effect; it is reported unadjusted, because this cell is *not* in the Benjamini–Hochberg family of the 31 intervention cells and is never pooled with them (different objective, horizon and dataset family). Two of the three degradation clauses fail as well: the gate does not clear 0.20 and full fine-tuning does not hurt. Early stopping did engage, at epochs 7, 7 and 9 under B against patience breaks at 12 to 14, so no run selected the checkpoint it started from. The encoder does move under B, by an $\ell_2$ of 1.4 to 1.8 at CKA 0.994 to 0.997, small, and now measured on windows no selection decision was made on. Under D, CKA is 1 and the drift is exactly zero because condition D freezes `t5.encoder.parameters()` for this backbone, which includes the token embedding the encoder reads through; Moirai's condition D leaves its input projection trainable, which is why it shows CKA below 1 (Appendix A6).

**What the cell shows.** This encoder moves very little under fine-tuning on this dataset, at CKA $\geq$0.994, where Moirai-Base on ETTh2 restructures to CKA 0.396 at $h$=192 (Table 24). That is two observations on two backbones, not a law.

## S8 The Baseline Ladder on Other Window Sets

### S8.1 The held-out windows (the retrospective variant)

**A denominator has to clear a floor before the ratio means anything.** If $\text{MSE}_{\text{base}}$ exceeds the training mean's, the gate is dividing by the error of an estimator worse than answering "the average", and a checkpoint can clear the threshold without beating anything, which is precisely what the trend extrapolation did. The fitted map is above the floor on **all 32 cells**, so it is never an inadmissible denominator. Boosted trees fall below it on 11 cells and seasonal naive on 6. For the trees the mechanism is that they cannot extrapolate beyond their training range, and ETTh2's test split is well outside it: the training-mean predictor itself scores 2.06 there at $h$=192 against a normalised variance of 1, and the trees score 4.82. Persistence falls below the floor on 17 and the MLP on 2. Counting

Table S8: **Gate $V_b$ under eight denominators and a floor, on one numerator, over the *held-out* windows.** The retrospective variant, retained because the prospective predictor of §4 was frozen against it. Bold clears 0.20. A dagger marks a denominator worse than the training-mean predictor on that cell: the ratio is still arithmetic, but it is not evidence of a pre-trained advantage, so daggered cells are excluded from the "and admissible" tally. The training-mean column is the floor, not a candidate estimator. The last row applies the three-clause degradation definition of §3 with clause (i) taken from each estimator and clauses (ii) and (iii) unchanged, over the 31 cells with a paired frozen-encoder run. Generated by `scripts/gate_baseline_sensitivity.py`.

| Cell | Fitted ridge (ours) | Training mean | Persist-ence | Seasonal naive | Per-feature AR | DLinear (shared) | Tuned ridge | MLP (128) | Boosted trees |
|---|---|---|---|---|---|---|---|---|---|
| Moirai-S / ETTh2 ($h$=192) | **+0.464** | **+0.845** | −0.063 | −0.267 | −0.283 | −0.738 | **+0.422** | **+0.296** | +0.934† |
| Moirai-S / ETTh2 ($h$=96) | **+0.400** | **+0.813** | −0.223 | −0.215 | −0.843 | −0.889 | **+0.374** | **+0.406** | **+0.806** |
| Moirai-B / ETTh2 ($h$=96) | **+0.361** | **+0.801** | −0.303 | −0.295 | −0.965 | −1.014 | **+0.333** | **+0.367** | **+0.793** |
| Moirai-B / ETTh2 ($h$=192) | **+0.275** | **+0.791** | −0.438 | −0.714 | −0.735 | −1.350 | **+0.218** | +0.048 | +0.911† |
| Moirai-L / ETTh2 ($h$=96) | **+0.219** | **+0.757** | −0.592 | −0.582 | −1.400 | −1.460 | +0.185 | **+0.227** | **+0.747** |
| TimesFM-2.5 / Electricity ($h$=24) | +0.132 | +0.179 | +0.254† | +0.301† | +0.132 | +0.132 | +0.132 | +0.186† | +0.146 |
| Moirai-S / ETTm2 ($h$=192) | +0.108 | **+0.830** | +0.101 | +0.058 | **+0.350** | +0.123 | +0.119 | **+0.545** | **+0.826** |
| Moirai-B / ETTm2 ($h$=192) | +0.083 | **+0.825** | +0.076 | +0.032 | **+0.332** | +0.099 | +0.095 | **+0.532** | **+0.821** |
| Chronos-T5-S / Electricity ($h$=24) | +0.048 | +0.100 | +0.182† | +0.233† | +0.048 | +0.048 | +0.048 | +0.107† | +0.064 |
| Moirai-S / Electricity7 ($h$=192) | −0.047 | **+0.662** | **+0.625** | −0.035 | −0.516 | −0.440 | −0.071 | −0.367 | +0.893† |
| Moirai-B / ETTm2 ($h$=96) | −0.057 | **+0.860** | +0.165 | +0.047 | +0.014 | −0.193 | −0.067 | +0.193 | **+0.855** |
| TimesFM-2.5 / Weather ($h$=24) | −0.077 | **+0.494** | +0.535† | **+0.209** | −0.077 | −0.077 | −0.077 | −0.002 | +0.018 |
| Moirai-S / Electricity7 ($h$=96) | −0.085 | **+0.701** | **+0.649** | +0.093 | −0.404 | −0.400 | −0.102 | −0.337 | +0.856† |
| TimesFM-2.5 / ETTh1 ($h$=24) | −0.101 | **+0.424** | +0.124 | **+0.291** | −0.101 | −0.101 | −0.091 | +0.005 | −0.005 |
| Chronos-T5-S / Weather ($h$=24) | −0.103 | **+0.481** | +0.523† | +0.190 | −0.103 | −0.103 | −0.103 | −0.027 | −0.006 |
| Moirai-S / ETTm2 ($h$=96) | −0.110 | **+0.853** | +0.124 | −0.000 | −0.035 | −0.252 | −0.121 | +0.153 | **+0.848** |
| Chronos-T5-S / ETTh1 ($h$=24) | −0.123 | **+0.412** | +0.107 | **+0.278** | −0.123 | −0.123 | −0.112 | −0.014 | −0.024 |
| Moirai-S / ETTh1 ($h$=96) | −0.243 | **+0.472** | +0.570† | **+0.216** | −0.362 | −0.242 | −0.297 | −0.147 | +0.621† |
| TimesFM-2.5 / ETTh2 ($h$=24) | −0.287 | **+0.475** | +0.527† | +0.021 | −0.287 | −0.287 | −0.238 | −0.108 | −0.131 |
| Chronos-T5-S / ETTh2 ($h$=24) | −0.325 | **+0.460** | +0.513† | −0.008 | −0.325 | −0.325 | −0.274 | −0.141 | −0.164 |
| Moirai-S / ETTh1 ($h$=192) | −0.331 | **+0.326** | +0.406† | +0.083 | −0.410 | −0.351 | −0.382 | −0.227 | +0.758† |
| Moirai-L / ETTh1 ($h$=96) | −0.454 | **+0.382** | +0.497† | +0.083 | −0.594 | −0.453 | −0.517 | −0.342 | +0.556† |
| Moirai-B / ETTh1 ($h$=96) | −0.458 | **+0.380** | +0.496† | +0.080 | −0.599 | −0.458 | −0.522 | −0.346 | +0.555† |
| Moirai-B / ETTh1 ($h$=192) | −0.501 | **+0.239** | +0.330† | −0.034 | −0.590 | −0.524 | −0.559 | −0.383 | +0.727† |
| Moirai-L / Weather ($h$=96) | −0.621 | **+0.204** | +0.248† | +0.143 | −0.531 | −0.437 | −0.611 | −0.611 | +0.055 |
| Moirai-S / Weather ($h$=192) | −0.639 | −0.037 | +0.276† | +0.163† | −0.775 | −0.670 | −0.639 | −0.648 | +0.600† |
| Moirai-S / Weather ($h$=96) | −0.867 | +0.084 | +0.134† | +0.014 | −0.763 | −0.655 | −0.855 | −0.855 | −0.088 |
| TimesFM-2.5 / ETTm2 ($h$=24) | −0.945 | **+0.576** | +0.136 | +0.631† | −0.945 | −0.945 | −0.901 | −0.861 | −0.531 |
| Moirai-B / Weather ($h$=96) | −1.029 | +0.004 | +0.059† | −0.072 | −0.916 | −0.799 | −1.016 | −1.016 | −0.183 |
| Chronos-T5-S / ETTm2 ($h$=24) | −1.281 | **+0.503** | −0.013 | +0.567† | −1.281 | −1.281 | −1.229 | −1.182 | −0.795 |
| Moirai-B / Weather ($h$=192) | −1.324 | −0.470 | −0.027† | −0.187† | −1.517 | −1.368 | −1.323 | −1.337 | +0.433† |
| Moirai-S / ILI ($h$=24) | −1.625 | −0.304 | +0.027† | −0.786 | −1.625 | −1.625 | −1.638 | −1.301 | −1.563 |
| *Clearing* 0.20 | 5/32 | 25/32 | 14/32 | 8/32 | 2/32 | 0/32 | 4/32 | 6/32 | 18/32 |
| *and admissible* | 5/32 | 25/32 | 2/32 | 4/32 | 2/32 | 0/32 | 4/32 | 6/32 | 7/32 |
| *Below the floor* | 0/32 | 0/32 | 17/32 | 6/32 | 0/32 | 0/32 | 0/32 | 2/32 | 11/32 |
| *Degradation cells* | 0/31 | 6/31 | 0/31 | 1/31 | 0/31 | 0/31 | 0/31 | 0/31 | 0/31 |

only admissible passes, the eight rungs give **5, 2, 4, 2, 0, 4, 6 and 7 of 32** in ladder order; the widest is boosted trees, which adds the four ETTm2 cells, and the narrowest is DLinear, which admits **none**.

**The negative headline strengthens. The identity of the five survivors does not.** No estimator admits many more cells than the fitted map and two admit far fewer, so "on most cells there is no demonstrated advantage" is not an artefact of one estimator family. But the passing *sets* are nearly disjoint. All five ETTh2 survivors fail against seasonal naive ($-0.215$ to $-0.714$), against per-feature AR ($-0.283$ to $-1.400$) and against DLinear ($-0.738$ to $-1.460$), which is the rung that beats the pre-trained model everywhere. Tuned ridge and the MLP each reproduce four of the five and not the same four: tuned ridge loses Moirai-Large/ETTh2 $h$=96 at $+0.185$, the MLP loses Moirai-Base/ETTh2 $h$=192 at $+0.048$. The denominators say why: on ETTh2 $h$=192 the univariate AR map scores 0.249 where the paper's multivariate map scores 0.595, so the cross-channel terms *hurt* there, and "Moirai beats a target-data regression on ETTh2" is in part a statement about a regression with 672 free inputs. The four cells seasonal naive admits are a different set again: Chronos/ETTh1, TimesFM/ETTh1, TimesFM/Weather and Moirai-Small/ETTh1 $h$=96. The instability of the passing

*set* under a change of rung is the same on both splits; it is why the body scopes its drift analyses by the union rather than by any one rung.

**Does the zero survive? Under seven of the eight rungs, yes.** With clause (i) taken from each estimator, the definition returns **zero cells under the fitted map, persistence, per-feature AR, DLinear, tuned ridge, the MLP and boosted trees, and one under seasonal naive**: Moirai-Small/ETTh1 $h{=}96$, which reads $+0.216$ there against $-0.243$ against the fitted map and then satisfies both remaining clauses unanimously, forg.$_{\mathrm{B}}{=}+18.5\%$ in 5/5 seeds, forg.$_{\mathrm{D}}{=}-3.5\%$ in 5/5, B$-$D$=+22.1$ at CKA $0.807$. We report it rather than absorbing it: against one of the eight rungs this paper's own definition has a positive instance.

Two things about that instance decide how much weight it carries. It clears $0.216$ against a threshold of $0.20$ and fails at $0.25$. And it arises from a *weaker* admissible denominator, not a stronger one: on ETTh1 $h{=}96$ seasonal naive scores $0.657$ where the fitted map scores $0.415$. The direction of the sensitivity is therefore that loosening the baseline manufactures degradation cells, the same mechanism that produces the unfitted baseline's eight, rather than a stronger baseline revealing damage we missed. The floor column is that trend taken to its limit: under the training-mean denominator the count is **six**, and the six are Moirai-Small/ETTh1 at both horizons, Moirai-Base/ETTh1 at both, Moirai-Large/ETTh1 $h{=}96$ and Moirai-Large/Weather $h{=}96$, all among the unfitted baseline's eight: those instances need a denominator no stronger than a constant.

**The ladder read as a graded axis, and why the headline is a minimum.** A pass count per rung still treats the gate as binary. The graded reading assigns each cell one number, $V_{\text{best}}$, measured against the *strongest* admissible rival it faces. That is a **minimum** over rungs, not a maximum: a weaker denominator inflates $V_b$, so the strongest rival is the one that leaves the smallest score. Read that way, **no cell of the 32 clears** $0.20$ **against every admissible rung**, and 15 clear it against at least one. The two counts move in opposite directions as the ladder grows (the first can only fall, the second can only rise), so only the first is a claim, and adding a ninth rung could not undo it. The five ETTh2 survivors have $V_{\text{best}}$ between $-0.738$ and $-1.460$, every one of them against DLinear. Both readings agree with the selection-split ones above (no cell clears every admissible rung on either split), so the graded conclusion does not depend on which split supplies the axis, and the value axis §6 plots is the selection-split one (`results/gate_baselines_val.json`) because its $x$ and $y$ must not share windows. Each file stores $V_{\text{best}}$ per cell alongside which rung supplied it.

### S8.2 A THIRD WINDOW SET: THE LADDER ON THE TRAIN TAIL

The selection-split gate meets the protocol a reviewer would ask for (the gate's windows are disjoint from the outcome's), but not its residual: the selection windows also drove early stopping and checkpoint choice, so the outcome depends on them indirectly. So we scored the same nine denominators a third time, on the last $20\%$ of each cell's *training* region, held back by index rather than inferred from which windows training happened to skip. Every rung's denominator is fitted on the earlier part of train only and the numerator is a zero-shot pass, so neither arm of the gate has seen these windows. The precise claim is that they are used for neither checkpoint selection nor outcome scoring; it is *not* that they were never seen, because the condition-B and -D runs subsampled their training windows from the whole train region including the tail. That does not touch the gate (whose numerator is the untrained checkpoint), but it is the difference between this arm and a genuinely fresh dataset. The arm covers the 21 Moirai cells (`scripts/gate_thirdsplit.py`, `results/gate_baselines_traintail.json`); Table S9 counts all three window sets over exactly those 21, so its columns differ by windows and not by coverage.

**The bound is confirmed on all three window sets; the survivor set is not.** No cell clears $0.20$ against every admissible rung on any of them: **0, 0 and 0 of 21**. The fitted rung, which is where the paper's seven survivors come from, does not transfer: it admits **7 cells on the selection windows and 3 on the train tail, with only 2 in both** (Moirai-Base/ETTh2 $h{=}192$ and Moirai-Small/ETTh2 $h{=}192$), and the train tail admits one cell the selection windows reject, Moirai-Small/ETTh1 $h{=}192$. The union moves the other way, 14 to 19, because the training-mean floor is so much worse on the tail that more rungs clear admissibility there at all.

**What moves is the denominator, not the checkpoint.** On Moirai-Base/ETTh2 $h{=}96$ the fitted ridge scores $1.012$ on the selection windows and $0.276$ on the train tail ($3.7\times$ better), while zero-shot barely moves, $0.225$ to $0.306$. The ridge is fitted on the training region, so its error grows once it

Table S9: **The eight-rung ladder on three window sets.** Cells each rung admits: its denominator is at least as good as the training-mean floor on that cell *and* the cell's $V_b$ clears $0.20$. Every column is out of the same 21 Moirai cells (the ones the train-tail arm covers), so the counts are comparable across columns and are not the 31- and 32-cell counts quoted elsewhere. *Selection* is the paper's primary, *held-out* the retrospective variant, and *train tail* the last $20\%$ of the training region, used for neither checkpoint selection nor outcome scoring. The bottom row is the paper's conservative bound, and it is zero on all three.

| Rung (clause (i)'s denominator) | Selection | Held-out | Train tail |
|---|---|---|---|
| Fitted ridge (ours) | 7 | 5 | 3 |
| Training mean (floor) | 16 | 17 | 20 |
| Persistence | 6 | 2 | 12 |
| Seasonal naive | 0 | 1 | 0 |
| Per-feature AR | 6 | 2 | 0 |
| DLinear (shared) | 4 | 0 | 0 |
| Tuned ridge | 7 | 4 | 2 |
| MLP (128) | 9 | 6 | 3 |
| Boosted trees | 5 | 7 | 9 |
| Clears $0.20$ against *at least one* admissible rung | 14 | 12 | 19 |
| Clears $0.20$ against *every* admissible rung | **0** | **0** | **0** |

is evaluated away from it, and that growth is most of what the ETTh2 survivors' margin measures. Beside its own fit region the ridge is much harder to beat and three cells retain a pre-trained advantage. This is the effect of Appendix A12 in a second guise: there the gate moves because the denominator is the wrong estimator, here because it is evaluated on different windows, and in both cases the quantity that moved was the baseline rather than the pre-trained model. A gate-pass count is a statement about a baseline *and* a window set, and this paper's counts should be read as such. What survives all three is the negative claim, which is the one the paper's conclusions rest on.

### S8.3 Matching the lookback: is the verdict an artifact of the context asymmetry?

Moirai is evaluated with $96{+}h$ steps of history (Figure S1) while the baseline is given the last $96$, and §4 argues that this favours the pre-trained model. An argument is not a measurement, so here is the measurement: the same ridge map, the same windows, the same fitting region, the same normalisation and the same $\lambda$, given all $96{+}h$ available steps instead of the last $96$. Only the integer passed to the fit changes, so the two columns of Tables S10 and S11 differ in the baseline's information set and in nothing else; the numerator is the same stored $\mathrm{MSE}_{\mathrm{ZS}}$, and the `fitted` column reproduces the published denominator on all 21 Moirai cells of each split to a relative tolerance of $10^{-6}$ before anything is reported.

**The effect is small and its sign depends on the split, because a matched map is not automatically a stronger one.** On the selection windows (the split every admission decision is made on), the matched map is the stronger of the two on **12 of 21** cells, and the count clearing $0.20$ falls from **7 to 6**: **Moirai-Base/ETTm2** $h{=}192$ is the one cell that flips, from $+0.211$ to $+0.125$. On the held-out windows the direction reverses. There the matched map is stronger on only **7 of 21**, the count rises from **5 to 7**, and the two cells that flip do so *into* the survivor set (Moirai-Small and Moirai-Base on ETTm2 at $h{=}192$). The reason is visible in the two MSE columns: the matched map estimates $(96{+}h){\times}F$ inputs from the same training windows, which fits the training region better and generalises to the held-out region worse. So giving the baseline the model's full history moves the count by one or two cells in whichever direction the extra parameters happen to pay off, and no reading in this paper turns on either move.

**Why this is reported as its own axis rather than as a ninth rung, and what the separation does not hide.** The ladder above varies the estimator *family* at a fixed lookback, and its rungs feed the graded $V_{\mathrm{best}}$ that §6 uses as a covariate; a rung that varies the *information set* instead would move that minimum for reasons belonging to a different question. The separation protects no count, and

we checked rather than asserted that: folding the matched column into the graded minimum leaves the number of cells clearing $0.20$ against every admissible rung at **0 of 21 on both splits**, changes **no** cell's graded verdict, and makes the matched map the strongest admissible rival on **0** cells on the selection split and **3** on the held-out split.

Table S10: **The gate with the baseline given Moirai's full $96{+}h$ history, on the *selection* windows.** The same fitted ridge in both columns; only how many of the available past steps it may use differs. MSE ratio is $\mathrm{MSE}_{96+h}/\mathrm{MSE}_{96}$, so below 1 means the extra history made the baseline stronger. Bold clears $0.20$. Generated by `scripts/matched_lookback_gate.py -split val`.

| Cell | $V_{\text{ridge}}$ (96) | $V_{\text{ridge}}$ ($96{+}h$) | $\Delta$ | MSE ratio | Verdict |
|---|---|---|---|---|---|
| Moirai-S / ETTh2 ($h$=192) | $\mathbf{+0.932}$ | $\mathbf{+0.903}$ | $-0.029$ | 0.699 | pass (both) |
| Moirai-B / ETTh2 ($h$=192) | $\mathbf{+0.925}$ | $\mathbf{+0.893}$ | $-0.032$ | 0.699 | pass (both) |
| Moirai-L / ETTh2 ($h$=96) | $\mathbf{+0.875}$ | $\mathbf{+0.858}$ | $-0.017$ | 0.879 | pass (both) |
| Moirai-S / ETTh2 ($h$=96) | $\mathbf{+0.874}$ | $\mathbf{+0.856}$ | $-0.017$ | 0.879 | pass (both) |
| Moirai-B / ETTh2 ($h$=96) | $\mathbf{+0.777}$ | $\mathbf{+0.747}$ | $-0.031$ | 0.879 | pass (both) |
| Moirai-S / ETTm2 ($h$=192) | $\mathbf{+0.352}$ | $\mathbf{+0.281}$ | $-0.071$ | 0.901 | pass (both) |
| Moirai-B / ETTm2 ($h$=192) | $\mathbf{+0.211}$ | $+0.125$ | $-0.086$ | 0.901 | pass → **fail** |
| Moirai-S / ETTm2 ($h$=96) | $-0.164$ | $-0.233$ | $-0.070$ | 0.944 | fail (both) |
| Moirai-S / ETTh1 ($h$=96) | $-0.177$ | $-0.145$ | $+0.032$ | 1.028 | fail (both) |
| Moirai-L / ETTh1 ($h$=96) | $-0.251$ | $-0.217$ | $+0.034$ | 1.028 | fail (both) |
| Moirai-B / ETTm2 ($h$=96) | $-0.257$ | $-0.332$ | $-0.075$ | 0.944 | fail (both) |
| Moirai-B / ETTh1 ($h$=96) | $-0.312$ | $-0.276$ | $+0.036$ | 1.028 | fail (both) |
| Moirai-S / ETTh1 ($h$=192) | $-0.347$ | $-0.368$ | $-0.021$ | 0.984 | fail (both) |
| Moirai-S / Electricity7 ($h$=192) | $-0.457$ | $-0.383$ | $+0.074$ | 1.054 | fail (both) |
| Moirai-S / Weather ($h$=192) | $-0.506$ | $-0.453$ | $+0.053$ | 1.037 | fail (both) |
| Moirai-S / Weather ($h$=96) | $-0.546$ | $-0.503$ | $+0.043$ | 1.029 | fail (both) |
| Moirai-B / ETTh1 ($h$=192) | $-0.600$ | $-0.625$ | $-0.025$ | 0.984 | fail (both) |
| Moirai-B / Weather ($h$=192) | $-0.632$ | $-0.574$ | $+0.057$ | 1.037 | fail (both) |
| Moirai-L / Weather ($h$=96) | $-0.639$ | $-0.593$ | $+0.046$ | 1.029 | fail (both) |
| Moirai-S / Electricity7 ($h$=96) | $-0.651$ | $-0.904$ | $-0.253$ | 0.867 | fail (both) |
| Moirai-B / Weather ($h$=96) | $-0.819$ | $-0.768$ | $+0.051$ | 1.029 | fail (both) |
| *Clearing* 0.20 | 7/21 | 6/21 | | 12/21 stronger | 1 lost, 0 gained |

### S8.4 THE NUMERATOR IS NOT DETERMINISTIC EITHER: THE EVALUATOR'S SAMPLING NOISE

Every denominator above is a fitted estimator, and the eight rungs ask whether the screen is a property of that choice. The *numerator* deserves the same question. Moirai's forecast is the median of 20 sampled paths, so a cell's zero-shot MSE is a measurement, not a constant: two runs of one cell on the same deterministic windows record two different values (Appendix S3). Each gate value in this paper is therefore one draw, and the width of that draw sets the resolution at which a $0.20$ operating point can be read at all.

We can measure it, because the records contain repeats. Across the 26 Moirai cell-scorings (the 21-cell retrospective grid and batch 3's five) there are 242 independent zero-shot measurements, every one of them the same frozen checkpoint on the same windows. Recomputing $R^2_{\text{task}}$ from each gives a per-cell spread of at most $0.0244$, on Moirai-Base/ETTh2 $h$=96; substituting the all-replicate mean for the registered denominator moves $R^2_{\text{task}}$ by at most $0.0038$ on batch 1, $0.0044$ on batch 2 and $0.0011$ on batch 3. (Batch 2 was abandoned before any outcome run: it contributed six zero-shot measurements, which enter batch 3's registered denominator, and no condition-B or condition-D record anywhere in this paper.)

**The comparison that matters is per cell, not global.** That largest spread, $0.0244$, is bigger than the tightest margin to the operating point anywhere ($0.0109$, Moirai-Base/ETTm2 $h$=192 at $R^2_{\text{task}}$=+0.2109), and pairing the two would say the screen sits inside its own noise. It does not: they are different cells, and the cell carrying the $0.0244$ sits $0.577$ from $0.20$, where noise of any size cannot change a call. Sampling noise on a cell can only threaten that cell's own call, so margin and noise must be taken together per cell. The minimum of that ratio over all 26 cell-scorings is $9.2\times$, at the tightest cell; and directly, **no cell's** $0.20$ **call flips under any single replicate substitution**, which

Table S11: **The same comparison on the *held-out* windows.** The direction reverses: the matched map is stronger on 7 of 21 cells here against 12 of 21 on the selection windows, because the larger design matrix generalises away from its fit region worse. Generated by `scripts/matched_lookback_gate.py -split test`.

| Cell | $V_{\text{ridge}}$ (96) | $V_{\text{ridge}}$ (96+$h$) | $\Delta$ | MSE ratio | Verdict |
|---|---|---|---|---|---|
| Moirai-S / ETTh2 ($h$=192) | $+\mathbf{0.464}$ | $+\mathbf{0.624}$ | $+0.161$ | 1.428 | pass (both) |
| Moirai-S / ETTh2 ($h$=96) | $+\mathbf{0.400}$ | $+\mathbf{0.389}$ | $-0.011$ | 0.982 | pass (both) |
| Moirai-B / ETTh2 ($h$=96) | $+\mathbf{0.361}$ | $+\mathbf{0.349}$ | $-0.012$ | 0.982 | pass (both) |
| Moirai-B / ETTh2 ($h$=192) | $+\mathbf{0.275}$ | $+\mathbf{0.492}$ | $+0.217$ | 1.428 | pass (both) |
| Moirai-L / ETTh2 ($h$=96) | $+\mathbf{0.219}$ | $+\mathbf{0.205}$ | $-0.014$ | 0.982 | pass (both) |
| Moirai-S / ETTm2 ($h$=192) | $+0.108$ | $+\mathbf{0.242}$ | $+0.134$ | 1.177 | fail → **pass** |
| Moirai-B / ETTm2 ($h$=192) | $+0.083$ | $+\mathbf{0.221}$ | $+0.138$ | 1.177 | fail → **pass** |
| Moirai-S / Electricity7 ($h$=192) | $-0.047$ | $-0.005$ | $+0.042$ | 1.042 | fail (both) |
| Moirai-B / ETTm2 ($h$=96) | $-0.057$ | $+0.081$ | $+0.138$ | 1.150 | fail (both) |
| Moirai-S / Electricity7 ($h$=96) | $-0.085$ | $-0.150$ | $-0.065$ | 0.944 | fail (both) |
| Moirai-S / ETTm2 ($h$=96) | $-0.110$ | $+0.035$ | $+0.145$ | 1.150 | fail (both) |
| Moirai-S / ETTh1 ($h$=96) | $-0.243$ | $-0.066$ | $+0.176$ | 1.165 | fail (both) |
| Moirai-S / ETTh1 ($h$=192) | $-0.331$ | $-0.179$ | $+0.152$ | 1.129 | fail (both) |
| Moirai-L / ETTh1 ($h$=96) | $-0.454$ | $-0.248$ | $+0.206$ | 1.165 | fail (both) |
| Moirai-B / ETTh1 ($h$=96) | $-0.458$ | $-0.251$ | $+0.207$ | 1.165 | fail (both) |
| Moirai-B / ETTh1 ($h$=192) | $-0.501$ | $-0.330$ | $+0.171$ | 1.129 | fail (both) |
| Moirai-L / Weather ($h$=96) | $-0.621$ | $-0.622$ | $-0.001$ | 0.999 | fail (both) |
| Moirai-S / Weather ($h$=192) | $-0.639$ | $-0.522$ | $+0.117$ | 1.077 | fail (both) |
| Moirai-S / Weather ($h$=96) | $-0.867$ | $-0.868$ | $-0.001$ | 0.999 | fail (both) |
| Moirai-B / Weather ($h$=96) | $-1.029$ | $-1.030$ | $-0.002$ | 0.999 | fail (both) |
| Moirai-B / Weather ($h$=192) | $-1.324$ | $-1.157$ | $+0.166$ | 1.077 | fail (both) |
| *Clearing* 0.20 | 5/21 | 7/21 | | 7/21 stronger | 0 lost, 2 gained |

the emitter asserts rather than reports, so a future record that broke it would fail the re-derivation instead of shipping.

Two consequences we accept. First, the screen's resolution is roughly $10^{-2}$ in $R^2_{\text{task}}$, so Moirai-Base/ETTm2 $h$=192 is a cell whose admission we would not defend on a remeasurement at a 0.21 threshold, which is why Table S2 sweeps the threshold on the pre-registered cells rather than defending one number. Second, this noise is why the pre-registered roots are read `condition_A` only: averaging the outcome runs' re-measurements into a registered denominator moves the predictor after its outcomes exist, and at $\sim 10^{-3}$ it does so far too quietly to be noticed downstream. The code asserts this structurally: a run fails if an outcome file is ever read as a registered reference.

## S9 Is the Zero a Property of the Runs or of the Definition?

The paper's headline is that no cell meets its definition of damaging a demonstrated pre-trained capability, and the first fair objection is that the definition does the work. It requires condition-B forgetting positive in *every* seed and condition-D forgetting negative in *every* seed; unanimity over ten seeds is a demanding bar, and if a simple majority produced five degradation cells the negative result would be a property of the rule rather than of the runs. That objection is answerable from the stored per-seed sign counts with no new runs.

Table S12 walks a ladder of definitions from the weakest defensible one (the two means alone, with seed signs unchecked) up to ours, crossed with both gate thresholds so the sensitivity to *both* free parameters is one table rather than two. Two exclusions apply identically at every rung, exactly as `degradation_cells()` applies them: cells with no zero-shot test reference cannot be scored at all (there are none), and the five Chronos cells are excluded because a head/decoder mismatch makes their per-condition forgetting non-comparable (limitation iv). Admitting them at the relaxed rungs would make loosening the rule appear to find degradation cells when what it found is a measurement already known to be invalid. Both exclusions are properties of the measurement rather than of the outcome, so applying them uniformly is not a selection on the result. That leaves 26 scorable cells, of

which 7 clear a gate of $0.10$ and 7 clear $0.20$: under the fitted baseline the two thresholds select the same set, so the gate column is a sensitivity check that happens to move nothing rather than a second free parameter.

Table S12: Degradation cells admitted under eight definitions and two gate thresholds. Cells are named, not just counted. Under the fitted gate no cell is admitted at $0.10$ alone, so the two columns agree throughout. The excluded confounded cells are Chronos-T5-Small on ETTh1, ETTh2, ETTm2, Weather and Electricity. Emitted by `scripts/degradation_sensitivity.py`.

| | Cells admitted | | |
|---|---|---|---|
| Definition of a degradation cell | Gate $\geq 0.10$ | Gate $\geq 0.20$ | Which cells |
| Means only (seed signs unchecked) | 1 | 1 | S/ETTh2 $h192$ |
| Means, $+$ $\geq$ half the seeds agree | 1 | 1 | S/ETTh2 $h192$ |
| Means, $+$ $\geq 70\%$ of seeds agree | 0 | 0 | — |
| Means, $+$ $\geq 80\%$ of seeds agree | 0 | 0 | — |
| **Means, $+$ every seed agrees (ours)** | 0 | 0 | — |
| Means, $+$ one-sided $t$ on $\text{forg}_\text{B}$ at $\alpha$=0.05 | 0 | 0 | — |
| Seed counts only, $\geq$ half agree | 2 | 2 | S/ETTh2 $h192$, S/ETTm2 $h192$ |
| Seed counts only, every seed agrees | 0 | 0 | — |
| *Gate-passing cells screened* | 7 | 7 | |

**Unanimity is not what produces the zero.** At the paper's gate threshold the largest count anywhere on the ladder is **two**, and it is produced by a single rung (seed counts only, half the seeds agreeing), which is the one rung that drops the means clause entirely. Of the six rungs that keep it, the *weakest*, the two means with seed signs unchecked, admits one, and requiring 70% of seeds, 80%, unanimity, or a one-sided $t$-test on $\text{forgetting}_\text{B}$ all give zero. Relaxing the gate to $0.10$ changes none of these counts, so 5 of the 8 rungs yield zero at both. There is no setting of these two parameters at which the paper's conclusion becomes "several cells degrade".

**The one cell that ever appears at a rung that keeps the means clause.** Moirai-Small on ETTh2 at $h$=192 has $\text{forgetting}_\text{B}$ $+1.99$ with an SEM of $3.32$ over ten seeds, a one-sided $p$ of $0.28$ against zero, and 5 of 10 seeds on each side. It is a coin flip, which is what the body already reports it as. Of the other six gate-passing cells, three are strongly *improved* by full fine-tuning rather than marginally harmed ($\text{forgetting}_\text{B}$ $-31.44\pm3.65$, $-36.35\pm2.80$ and $-42.42\pm0.47$ SEM, with every seed agreeing), and two more improve on the mean, both ETTm2 at $h$=192 ($-1.10\pm3.37$ and $-8.45\pm3.23$). The seventh, Moirai-Small/ETTh2 at $h$=96, does have a positive mean $\text{forgetting}_\text{B}$ of $+9.71\pm10.58$, but fails the means clause on the *other* side, for the reason the next paragraph gives.

**Means and seed counts can disagree in direction, and we report both families rather than absorb the disagreement.** Moirai-Small/ETTh2 at $h$=96 has a mean $\text{forgetting}_\text{D}$ of $+6.91$ while 9 of its 10 seeds are negative: a single outlier drags the mean across zero. A ladder that silently required the means clause at every rung would hide that, and "majority seed-consistency" is what the objection proposes as a *replacement* for unanimity, not as an extra hurdle on top of it. The last two rows therefore drop the means clause entirely, and the first of them is the only row on the whole ladder that admits two cells, which is how Moirai-Small/ETTm2 at $h$=192 becomes visible at all.

**Which definition the paper uses, and why.** We keep unanimity. The reason is not that it yields the fewest cells (at gate $0.20$ 5 of the 8 rungs also yield zero), but that a degradation claim is a claim that fine-tuning *reliably* destroys something, and a cell whose seeds disagree on the sign of the harm is not evidence of that. The cost of the choice is that it can discard a real but noisy effect, which is exactly what the $h$=192 cell would be if its true $\text{forgetting}_\text{B}$ were small and positive; that cell is therefore reported by name in the body rather than dropped silently.

Table S13: **The pre-registered seed top-up, beside the published seed counts.** The 7 cells clearing $V_{\text{ridge}} \geq 0.20$. 3 were topped up under `results/power_topup/preregistration_power.json`, which fixed which cells, which seeds and how many (26 runs) before any of those runs existed; all 26 completed. **No published number is replaced.** Each cell's *published* row is the analysis of Table 21, re-verified against `results/paired_inference.json` at emit time; its *augmented* row is a second computation with the added seeds admitted, on the same estimator and the same Benjamini–Hochberg correction over the same 31 cells. **One call changes**: Moirai-S / ETTm2 ($h$=192, $n$=1000) moves from inconclusive to freezing-decisive, first at $n$=12, with the rung below it at $q$=0.059. Across all 31 cells the augmented read changes that call and no other (published 2 freezing-better, 4 adaptation-better, 25 inconclusive; augmented 3, 4, 24). The 2 cells on the *uncorrected* tier were run to a budget below their BH-adjusted requirement and were registered as *expected to remain inconclusive*, which held for 1 of the 2 (Moirai-S / ETTh2 ($h$=192, $n$=500)) and failed for Moirai-S / ETTm2 ($h$=192, $n$=1000). A failed registered prediction is reported as one; the registration is not edited. The ladder behind "first at" admits each cell's added seeds in registered order with the other topped-up cells at their final $n$, so a change is attributable to that cell's own seeds and not to BH moving underneath it. The last column gives each cell's registered budget and the tier it came from (adj. = the BH-adjusted requirement, unc. = the uncorrected one), or the reason it was not topped up.

| Cell | $V_{\text{ridge}}$ | read | $n$ | $\Delta_{\text{enc}}$ | $q$ | call | top-up |
|---|---|---|---|---|---|---|---|
| Moirai-B / ETTh2 ($h$=192, $n$=1000) | +0.925 | published | 3 | −20.2±1.1 | 0.018 | adapt | not run: already decisive |
| Moirai-B / ETTh2 ($h$=96, $n$=1000) | +0.777 | published | 3 | −22.3±2.9 | 0.071 | incon. | |
| | | augmented | 5 | −16.4±4.7 | 0.077 | incon. | 3→5 (adj.) |
| Moirai-B / ETTm2 ($h$=192, $n$=1000) | +0.211 | published | 3 | +1.3±3.4 | 0.740 | incon. | not run: needs $n$=83 |
| Moirai-L / ETTh2 ($h$=96, $n$=500) | +0.875 | published | 3 | −20.8±1.4 | 0.022 | adapt | not run: already decisive |
| Moirai-S / ETTh2 ($h$=192, $n$=500) | +0.932 | published | 10 | +6.1±3.4 | 0.173 | incon. | |
| | | augmented | 14 | +6.9±2.5 | 0.059 | incon. | 10→14 (unc.) |
| Moirai-S / ETTh2 ($h$=96, $n$=500) | +0.874 | published | 10 | +2.8±3.7 | 0.522 | incon. | not run: needs $n$=71 |
| Moirai-S / ETTm2 ($h$=192, $n$=1000) | +0.352 | published | 5 | +4.7±3.2 | 0.306 | incon. | |
| | | augmented | 12 | +5.7±1.8 | 0.043 | freeze | 5→12 (unc.) |

## S10 A PRE-REGISTERED SEED TOP-UP

**A pre-registered seed top-up, and the one call it changed.** Five of the seven admitted cells are inconclusive, and Table 22 says four of those five cannot be resolved at any budget we can run. What the resolvable ones do with more seeds is therefore the only power question still open on this matrix, and an answer to it is worth something only if the budget was fixed before the seeds ran. `results/power_topup/preregistration_power.json` was committed before any added run existed and fixes the rule (top up a cell iff its call is inconclusive and at least one of its two required $n$ is within the feasibility cap of 20), the three cells that rule selects, their seeds, and the exact $n$ at which to stop. All **26** registered runs completed, and `scripts/emit_power_topup.py` exits non-zero if any one of them is missing, so Table S13 is evidence that the top-up finished rather than our word for it. The published column stands: every other number in this paper is at the published $n$, and `cell_matrix.topup_paths()` excludes those 26 records from every published read so that they cannot leak into one.

**One cell changes its call, and it is a cell our own registration predicted would not.** Moirai-Small/ETTm2 $h$=192 goes from inconclusive at $n$=5 ($\Delta_{\text{enc}}$=+4.7±3.2, $q$=0.306) to *freezing decisively better* at $n$=12 (+5.7±1.8, $q$=0.043), and the ladder puts the change at $n$=12 itself, its last registered seed, with the rung below it at $q$=0.059. The registration marked both uncorrected-tier cells *expected to remain inconclusive*, on the ground that they were being run to the uncorrected requirement while a decisive call needed 29 paired seeds. That expectation held for Moirai-Small/ETTh2 $h$=192 ($n$=14, $q$=0.059) and **failed here**. We report the failure and leave the registration file as it was written. **Why it failed is worth more than the fact that it did.** The 29 was computed at $\alpha/m$, which is Bonferroni; the cell's paired $t$-test $p$ at $n$=12 is 0.0098, the 7th smallest of the 31, so the threshold it actually faces is $7\alpha/m$=0.011 — seven times the 0.0016 its budget was sized against. Sizing a budget at $\alpha/m$ bounds the BH threshold from below and so the required $n$ from above,

exactly as Table 22's caption says it does, and this cell is what that conservatism looks like when it bites: re-estimated at the augmented $n$, its own adjusted requirement is 18, not 29.

**The adjusted-tier cell moved the other way, which is the same lesson.** Moirai-Base/ETTh2 $h{=}96$ was run to $n{=}5$, the budget estimated to make its call decisive, and it is *still* inconclusive: $q$ went from $0.071$ to $0.077$, because the two added seeds took the mean from $-22.3$ to $-16.4$ and the sd from $5.0$ to $10.4$. Every required $n$ in Table 22 holds the observed sd fixed, and this cell's sd was observed on three seeds. So the top-up returns a cell that resolved on a budget sized not to resolve and a cell that failed to resolve on a budget sized to resolve, and both are the same fact about post-hoc power at these $n$: the sd is the least certain quantity in the calculation, and it is the one the calculation treats as known.

**What the augmented read moves, and what it does not.** Across the matrix it changes that one call and no other, and no untopped cell's $q$ moves by as much as $0.019$; the three-way split goes from 2 freezing-better, 4 adaptation-better and 25 inconclusive to **3, 4 and 24**. The degradation definition of §3 returns **zero cells** under the augmented read as under the published one, in both the registered unanimity form and the interval form, since $\Delta_{\text{enc}}$ is not what either clause tests. What does move is the asymmetry stated above, that this matrix's screen admits cells where adaptation decisively wins and rejects the only cells where preservation decisively wins: at the augmented $n$ there is one *admitted* cell where freezing decisively wins, at a gate of $+0.352$. That is the one result in this paper that points the screen's way on the screen's own admitted set, and it came from seeds whose number was fixed before they ran; it is also a single cell at $q{=}0.043$ against a $0.05$ threshold whose neighbour misses the same threshold from the other side at $0.059$. The body names the exception where it states the split (§5.4) and leaves the arithmetic here, because every number there is at the published $n$; we do not restate it as support for the screen, which across 31 cells it is not.

Generated by `scripts/paired_inference.py` and, for the top-up, `scripts/emit_power_topup.py`.

## S11 The Graded Value Axis: Is the Dissociation Confined to Cells With Nothing to Lose?

The screen in the body is binary: a cell either clears $V_b$=0.20 or it does not. That discards the only interesting covariate, and it leaves open the single strongest objection to this paper's negative result: that drift fails to predict $\Delta_{\text{enc}}$ only *because* 16 of our 31 cells have no measured advantage to lose, so we are reading noise where there is nothing to read. If that were right, the drift-to-outcome relation would strengthen with pre-trained value, and our finding would be a statement about weak benchmarks rather than about the diagnostic. This appendix estimates that dependence two ways and reports an interval rather than a verdict.

**The axis.** Value here is the graded score of Appendix A11: $V_{\text{best}}$ against the *strongest admissible* rung of the eight-baseline ladder, which is a **minimum** over rungs, since $V_b = 1 - \text{MSE}_{\text{ZS}}/\text{MSE}_{\text{base}}$ and a weaker rival inflates it. It runs from $-1.638$ (Moirai-Small/ILI) to $+0.058$ (Moirai-Small/ETTh2 $h$=96) with mean $-0.610$, and its ceiling below 0.20 is the same fact as "no cell clears the gate against every admissible rung". A cell with no admissible rung at all is *dropped*, never scored as zero value: a missing denominator and a denominator of zero are different facts, and the second would place the cell at the origin of this very axis.

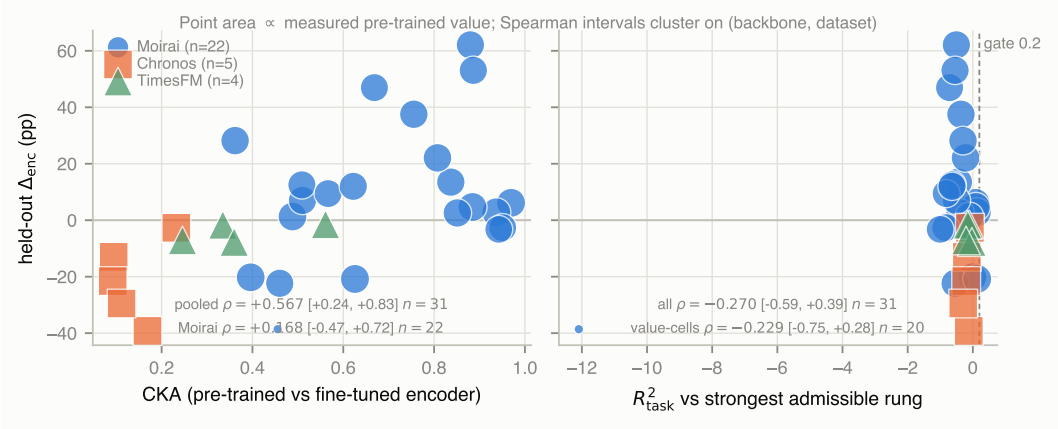



Figure S2: **Neither drift nor measured pre-trained value orders the encoder intervention.** Both panels share $y$= held-out $\Delta_{\text{enc}}$ (positive = freezing the encoder did better). Point area is proportional to the graded value score in *both* panels, so the left panel answers the objection visually: if drift predicted the outcome wherever there was something to lose, the large points would trend and the small ones would scatter. Backbone is encoded by colour *and* marker because CKA is not comparable across architectures, which is why the pooled left-panel interval must not be read as a within-backbone one. Drawn by `paper_8/fig_value_axis.py` from `results/value_axis.json`; the script asserts its own counts against that file and fails rather than drawing a figure that disagrees with this text.

**The rank answer: the relation is not stronger where there is more to lose.** Restricting to the 20 value-cells gives $\rho$=+0.567 with clustered CI $[+0.191, +0.889]$; the 11 cells with no admissible value give $\rho$=+0.709 with CI $[-0.077, +0.901]$ (Table S14). The restriction the objection asks for does not sharpen the relation, and the value-cell figure is not evidence that drift orders the outcome: it pools all three backbones, and within Moirai (the only arm with the cells to ask), the same statistic is $\rho$=+0.168 with an interval spanning zero. We do not test the difference of those two $\rho$s: a difference of rank statistics on 20 and 11 cells from 11 and 7 clusters has no usable sampling distribution, and reporting one would be the kind of number this paper is arguing against.

**The slope answer, where the dependence is a single coefficient.** On the slope scale the interaction is estimable, and it is $b_3$=−31.5 pp per unit CKA per unit value with CI $[-209.6, +166.2]$, against a CKA slope at mean value of +39.8 with CI $[-55.1, +179.4]$. Two things to say about it and no more. The point estimate has the *opposite* sign to the objection (drift orders the outcome slightly *less* well where there is more value), and the interval spans zero so widely that only an interaction several times larger than the CKA slope itself would have been visible at 15 clusters. **So this is not a refutation of**

Table S14: **The dependence, estimated two ways.** Rank rows are Spearman $\rho$ with 10,000 bootstrap draws over (backbone, dataset) clusters, seed 0. The slope rows fit all cells at once with backbone identity absorbed as a fixed effect and both regressors centred, so $b_1$ is the CKA slope *at mean value*; $b_3$ is the quantity at issue. Emitted by `scripts/value_axis.py -latex`, which delegates the slope fit to `scripts/cka_fixed_effects.py` rather than refitting it.

| Relation | $n$ | clusters | $\rho$ or $b$ | 95% CI over clusters |
|---|---|---|---|---|
| CKA vs $\Delta_{\text{enc}}$, the value-cells | 20 | 11 | $+0.567$ | $[+0.191, +0.889]$ excludes 0 |
| CKA vs $\Delta_{\text{enc}}$, cells with no admissible value | 11 | 7 | $+0.709$ | $[-0.077, +0.901]$ includes 0 |
| CKA vs $\Delta_{\text{enc}}$, within Moirai | 22 | 6 | $+0.168$ | $[-0.469, +0.717]$ includes 0 |
| CKA vs $\Delta_{\text{enc}}$, pooled (not a licensed reading) | 31 | 15 | $+0.567$ | $[+0.239, +0.828]$ excludes 0 |
| $V_{\text{best}}$ vs $\Delta_{\text{enc}}$, the value-cells | 20 | 11 | $-0.229$ | $[-0.755, +0.284]$ includes 0 |
| $V_{\text{best}}$ vs $\Delta_{\text{enc}}$, all cells | 31 | 15 | $-0.270$ | $[-0.592, +0.392]$ includes 0 |
| *Slope scale, all cells at once:* $\Delta_{enc} = b_0 + b_1\text{CKA} +$ *backbone FE* $+ b_2 v + b_3(\text{CKA}\times v)$, $v =$ *value* | | | | |
| $b_1$, CKA slope at mean value (pp per unit CKA) | 31 | 15 | $+39.8$ | $[-55.1, +179.4]$ includes 0 |
| $b_3$, CKA$\times$value interaction | 31 | 15 | $-31.5$ | $[-209.6, +166.2]$ includes 0 |

**the objection; it is the statement that these cells cannot separate it from no dependence at all.** That is a weaker claim than we would like to make, and it is the one the data support.

**And value does not order the outcome by itself.** The graded score against $\Delta_{\text{enc}}$ gives $\rho{=}-0.270$, CI $[-0.592, +0.392]$ across all 31 cells and $\rho{=}-0.229$, CI $[-0.755, +0.284]$ across the 20 value-cells, the same non-result as CKA, on the covariate that was supposed to rescue it. The right panel of Figure S2 is that row: every point lies left of the 0.20 line, and the cells that improve most under encoder adaptation are not the ones with the least to lose.

**Not an artefact of cells with nothing to lose.** Drift might fail to order $\Delta_{\text{enc}}$ only because 11 of the 31 cells beat no rung of the ladder; grading the screen rather than dichotomising it says otherwise: the relation is no stronger on the cells with a measured advantage than on the cells without, and the estimated interaction has the wrong sign for the objection and an interval far too wide to settle it.

## S12 Is the Reading Specific to CKA? Six Measures on 59 Encoders

The body reads drift through linear CKA, so a reader is entitled to ask whether "drift does not order the outcome" is a fact about representations or a fact about CKA. Appendix A15 gives the answer that covers the whole matrix: $\ell_2$ weight drift, a geometrically unrelated quantity defined in §3 alongside CKA, fails on the same 31 cells. This appendix gives the second answer, on geometry, and states up front what it cannot do.

**What it cannot do.** Only the $n{=}10$k runs retained encoder checkpoints. The 22-cell Moirai matrix is $n{=}500/1{,}000$ and kept none, so this battery *cannot* recompute the within-Moirai $\rho$ under a different measure; that would require retraining the matrix, on different hardware, which would perturb every number in the paper. What it can do is test measure-specificity on the 59 encoders that do exist: Moirai-Small at $h{=}96$, $n{=}10$k, on ETTh1, ETTh2 and ETTm2, under full fine-tuning and under two layer-unfreezing regimes (top 3 of 6, and all 6), in seven (dataset, run-set, regime) groups.

**Measured on the representations the paper's CKA column describes.** Each run's record stores `final_cka`, computed during training on the first 300 validation evaluation windows with the encoder output mean-pooled. The battery (`scripts/drift_metric_battery.py`) rebuilds exactly those windows (extended lookback $96{+}96$, raw values, the model normalising internally), and recomputes linear CKA as a self-check: it agrees with the stored value to four decimal places on all 59 encoders, so every other column is known to describe the same representations rather than a similar-looking set. The checkpoint load asserts that `load_state_dict(strict=False)` reports no missing and no unexpected keys, because a namespace mismatch loads nothing silently and returns CKA $\approx 1$, which reads as "no drift".

**The more invariant the measure, the less drift there is to see.** Linear CKA spans 0.194 to 0.915 across these encoders. SVCCA (invariant to any invertible linear map, so it asks whether the same information survives in a possibly rotated and rescaled basis) spans only 0.726 to 0.939, and

Table S15: **Six measures of representation change on 59 Moirai-Small encoders.** All computed on identical windows and identical pooling. "$\rho$ with CKA" is the rank correlation of each measure with linear CKA over the 59 encoders; the last two columns correlate each measure with forgetting, pooled and then within each of the six groups of at least five encoders. Forgetting here is the *selection-split* figure: these runs stored no zero-shot test MSE, so a held-out contrast cannot be reconstructed from them, and §5.3 shows same-split scoring can invert the sign of the intervention. This arm is therefore a comparison of measures, not a held-out claim. Emitted by `scripts/drift_metric_battery.py`; the pooled column mixes three datasets and is descriptive for the reason Appendix A15 gives about pooling.

| | over 59 encoders | | | | $\rho$ with forgetting | |
|---|---|---|---|---|---|---|
| Measure | min | max | mean | $\rho$ with CKA | pooled | per group |
| Linear CKA | $+0.194$ | $+0.915$ | $+0.588$ | $+1.000$ | $+0.350$ | $[-0.19, +0.60]$ |
| Linear CKA (token) | $+0.626$ | $+0.957$ | $+0.751$ | $+0.542$ | $-0.123$ | $[-0.33, +0.55]$ |
| RBF CKA | $+0.624$ | $+0.913$ | $+0.734$ | $+0.412$ | $+0.117$ | $[-0.36, +0.90]$ |
| SVCCA | $+0.726$ | $+0.939$ | $+0.776$ | $+0.553$ | $-0.038$ | $[-0.70, +0.35]$ |
| Procrustes sim. | $+0.715$ | $+0.947$ | $+0.818$ | $+0.675$ | $+0.127$ | $[-0.52, +0.80]$ |
| Mean cosine | $+0.660$ | $+0.990$ | $+0.840$ | $+0.073$ | $-0.287$ | $[-0.67, +0.60]$ |
| Eff. rank ratio | $+0.969$ | $+1.250$ | $+1.119$ | $-0.251$ | $+0.245$ | $[+0.00, +0.65]$ |

orthogonal-Procrustes similarity $0.715$ to $0.947$. The cell that CKA scores at $0.194$ is at $0.73$ on SVCCA. This is the concrete form of the objection limitation (7) concedes: a large part of what CKA reports as drift is a change of basis under which the subspace, and with it the linearly decodable content, is largely preserved. It is also why we do not claim CKA is invalid (it measures what it measures), only that a paper cannot infer damage from it.

**The measures do not order the encoders alike, so "drift" is not one quantity.** Rank correlations among them run from $+0.073$ (linear CKA against mean cosine, the least invariant measure in the battery) through $+0.412$ (linear CKA against RBF CKA) to $+0.845$ (RBF CKA against Procrustes). Choosing a similarity measure chooses an ordering of these encoders. We had expected the opposite (a family of near-clones), and report the disagreement because it makes the next paragraph the load-bearing one.

**No measure orders the outcome either.** Against selection-split forgetting, the pooled rank correlations run from $+0.350$ (linear CKA) to $-0.287$ (mean cosine), and the pooled figures are the flattering ones: within groups, where dataset identity is held fixed, every similarity measure changes sign across the six groups (linear CKA $[-0.19, +0.60]$, RBF CKA $[-0.36, +0.90]$, SVCCA $[-0.70, +0.35]$, Procrustes $[-0.52, +0.80]$, mean cosine $[-0.67, +0.60]$). Six measures spanning no invariance to full linear-map invariance therefore reproduce the body's reading rather than overturning it: the disagreement among them does not resolve into any one of them ordering the outcome. The single exception is not a similarity measure at all. The effective-rank ratio (how many directions the representation uses after fine-tuning against before) is above 1 on 55 of the 59 encoders ($0.969$–$1.250$, so fine-tuning generally *expands* the spectrum) and is the one column with no negative group value ($[+0.00, +0.65]$). At six groups, one backbone and one sample size we record that as a lead for future work, not a diagnostic; it is exactly the kind of pattern this paper's own prospective test (§4) was built to keep us honest about.

**What this licenses, and what it does not.** Read together, the six measures support the generalisation our title implies but the body does not claim: on this arm, *post-fine-tuning representation similarity should not be assumed to identify harmful encoder adaptation*, whichever similarity is chosen. What that reading cannot be is a claim about $\Delta_{\text{enc}}$: these 59 encoders are one backbone at one capacity, horizon and sample size, none of them among the 31 cells the intervention is measured on, and their outcome column is single-arm selection-split forgetting. A measure that failed here could still order a held-out intervention elsewhere, and we did not test one that does.

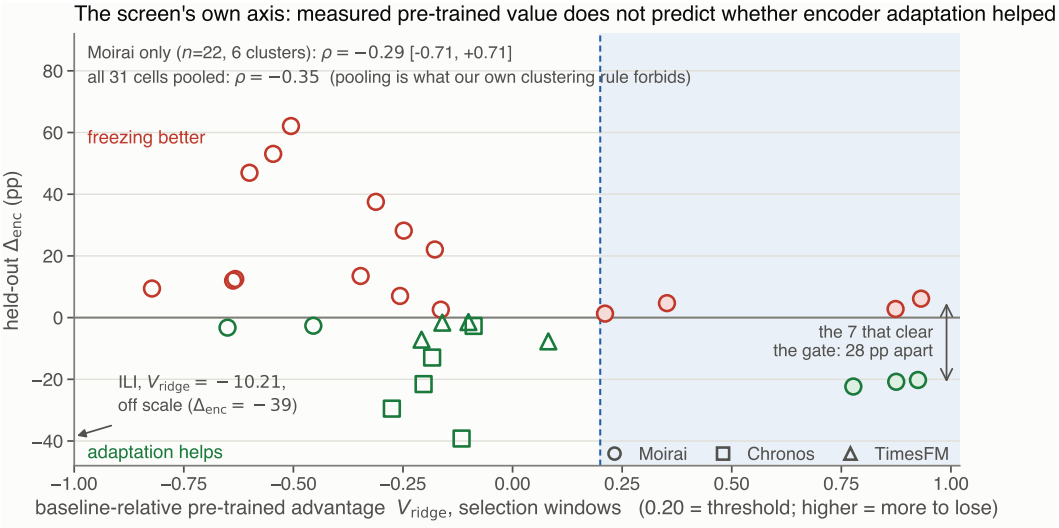



Figure S3: **The screen's own axis.** All 31 intervention cells as the selection-split gate against held-out $\Delta_{\text{enc}}$ (positive = freezing better). Filled markers clear $V_{\text{ridge}} \geq 0.20$; the shaded strip is the only region a cell could occupy and still satisfy clause (i) of the degradation definition, and most of the cells in it lie *below* zero, where letting the encoder adapt helped. The two axes are scored on disjoint window sets, so a cell's horizontal position is not a re-description of its vertical one. ILI is named on the frame rather than plotted: its gate is $-10.21$, and an axis wide enough to include it compresses the other 30 cells into a fifth of the panel.

## S13 What Predicts the Intervention, and Why None of It Is a Shortcut

If some quantity measurable from one run ordered B−D, a practitioner could skip the frozen control. We tested every candidate we had, within Moirai only, because it is the only arm with enough cells to ask (22). The statistic is again Spearman $\rho$ against held-out B−D with a 10,000-resample bootstrap CI at both resampling levels, from `scripts/clustered_inference.py`:

| Predictor (Moirai only) | $n$ | Cl. | $\rho$ | 95% CI over cells | 95% CI over clusters |
|---|---|---|---|---|---|
| $\text{forg}_\text{B}$ (condition B only) | 22 | 6 | $+0.894$ | $[+0.667, +0.981]$ excludes 0 | $[+0.594, +0.965]$ excludes 0 |
| $\text{forg}_\text{D}$ | 22 | 6 | $+0.073$ | $[-0.324, +0.429]$ includes 0 | $[-0.343, +0.412]$ includes 0 |
| Gate $V_{\text{ridge}}$ | 22 | 6 | $-0.293$ | $[-0.706, +0.188]$ includes 0 | $[-0.712, +0.709]$ includes 0 |
| CKA | 22 | 6 | $+0.168$ | $[-0.328, +0.598]$ includes 0 | $[-0.469, +0.717]$ includes 0 |
| $\ell_2$ weight drift | 21 | 5 | $+0.318$ | $[-0.122, +0.673]$ includes 0 | $[-0.329, +0.861]$ includes 0 |

Weight drift covers 21 of the 22 cells because ILI's run records retain no drift measurement. The gate row is the same figure reported in Appendix A12; it appears here beside its competitors. **These 22 cells come from six clusters**, and the gate row is the one place in the paper where that distinction changes a verdict: see Appendix A16.

**The $\text{forg}_\text{B}$ result is an identity, not a discovery.** B−D $\equiv \text{forg}_\text{B} - \text{forg}_\text{D}$ holds *exactly*, because both terms are percentage changes against the same zero-shot reference: over all cells the maximum deviation is $\max|(\text{forg}_\text{B} - \text{forg}_\text{D}) - (\text{B}-\text{D})| = 0.000$. The dispersion is then overwhelmingly on the B side, SD 26.2 against $\text{forg}_\text{D}$'s 8.1, so B−D is close to a rescaled copy of $\text{forg}_\text{B}$ and $\rho{=}{+}0.894$ is close to arithmetic rather than to a finding. It confers no ability to skip condition D: $\text{forg}_\text{B}$ exists only once condition B has been run, and at that point running D is one further run on the same cell. This is exactly the kind of correlation that looks like a shortcut and is not, and the near-zero $\text{forg}_\text{D}$ row is the reason: the rule works only because the frozen arm happens to be nearly constant on this backbone, which is an empirical property of these cells and not a theorem.

**The gate's continuous score points the wrong way, but only the sign is supported** (Figure S3). On the same 22 cells the gate correlates *negatively* with held-out B−D, $\rho{=}{-}0.293$ on the selection-

split gate this paper reports as primary. Neither resampling level supports an interval claim: cells give $[-0.706, +0.188]$ and the six clusters give $[-0.712, +0.709]$. On the retrospective test-side gate the same correlation is $\rho{=}-0.518$ with a cell-level CI of $[-0.840, -0.040]$ that *does* exclude zero, and a clustered CI of $[-0.810, +0.500]$ that does not, so we do not read it as excluding zero (Appendix A16); scoring the gate on the split that does not touch the outcome removes it independently, and the magnitude nearly halves between the two splits, which is itself the point of §4, that there is no stable ordering here to exploit. The mechanism is legible in Table 8: the cells with the largest measured pre-trained advantage are the Moirai/ETTh2 cells, and three of those five are the cells where full fine-tuning beats the frozen encoder by the most (B$-$D $-20.2$ to $-22.3$ pp), while the cells with the most negative gates (Weather and ETTh1) are where freezing wins by tens of points, up to $+62.1$ pp. A practitioner following "preserve where the gate fires" would therefore freeze precisely where adapting the encoder pays best. We report this at the strength it has: a within-Moirai point estimate on 22 retrospective cells from six series, with a clustered interval that spans zero. **The negative sign is a description of these cells, not an interval-supported claim, and the magnitude is not estimated to any useful precision.** The mechanism above is legible cell by cell in Table 8 and does not depend on the correlation surviving; that is why we still report the direction.

**The other two arms cannot adjudicate.** On TimesFM ($n{=}4$) the gate gives $\rho{=}+0.800$, CI $[+0.333, +1.000]$, forg$_\text{B}$ gives $+1.000$ and CKA $+0.400$, CI $[-0.778, +1.000]$; on Chronos ($n{=}5$) the gate gives $\rho{=}+0.500$, CI $[-0.667, +1.000]$, and forg$_\text{B}$ gives $-0.100$. These are reported for completeness and are uninformative at four and five cells: a bootstrap over four cells resamples only a few distinct rank patterns, so even the TimesFM gate interval, which nominally excludes zero, is an artefact of that granularity and we read nothing from it. Chronos's forg$_\text{B}$ carries a further defect and is flagged `forg_confounded` in the code: a head/decoder mismatch means its B-side number is not a clean per-condition forgetting measurement, so it is retained only to show where the forg$_\text{B}$ rule breaks and is never quoted as a forgetting value.

**Nothing here is a replacement diagnostic.** The honest summary of this appendix and of Appendix A15 together is that no observational quantity we measured orders the intervention in a usable direction: CKA and $\ell_2$ drift do not order it, the gate orders it *backwards*, and the one predictor that tracks it does so by identity and only after the run that would have told you the answer anyway. On the evidence in Table 8, the only reliable way to know whether encoder adaptation helped a given cell is to run both conditions on that cell.

## S14 The Same Cells Priced as a Decision, Post-Hoc

Every result above is a statement about an interval: a within-Moirai slope whose sign is not resolved, a $\Delta R^2$ that is negative but whose dispersion cannot be quoted. A reader may fairly reply that none of that is evidence *against* a relationship. So this subsection asks the question in the units a practitioner would act in, where the claim is falsifiable rather than merely unproven: what would following a CKA rule have been *worth*? **This analysis is post-hoc.** Unlike Table 26 nothing here was registered in advance; the policy family, the objective and the folds were all chosen after the outcomes were known, and the emitter (`scripts/cka_decision_value.py`) says so in its first paragraph.

**The construction.** A practitioner facing one cell picks full fine-tuning or a frozen encoder, and collects $\Delta_\text{enc}$ on every cell they freeze, so the realised saving of a policy against the default of always adapting is the mean over all 31 cells of $\Delta_\text{enc} \cdot \mathbf{1}[\text{freeze}]$, in percentage points of MSE. Every single-threshold CKA rule lies in one family, which puts six policies on one scale (Table S16). **The out-of-sample row is the only one a practitioner could have followed**: the cut *and* its orientation are searched on the 14 training clusters and applied unchanged to the held-out cluster, so every cell's decision is taken without seeing that cell. Candidate cuts are interior, so the search cannot quietly return *always freeze* and have it counted as CKA informing anything; both degenerate policies are priced as rows of their own.

**The number that does not favour our claim, first.** The best-orientation rule is worth $+7.47$ pp out of sample – not zero, and not what a reader would predict from a headline that drift carries no actionable information. Two things bound it. Its interval is $[-4, +16]$ pp and it is worth nothing or less in $25\%$ of cluster-bootstrap draws, so these cells do not *establish* a saving at all: this is a policy whose value is unresolved, exactly as the slope's sign is unresolved, and the honest reading is the same in both cases. And the orientation that earns it is **the inverse of the one the literature**

**prescribes**: it freezes the cells whose representations moved *least*. Freezing where drift is largest – what forgetting mitigation actually recommends – is worth $-6.97$ pp $[-10, +14]$, and is worth nothing or less in $80\%$ of draws. A practitioner who followed the prescription on these cells would have lost about seven points of MSE per cell relative to not measuring drift at all.

**What the other rows add.** Always freezing, ignoring CKA entirely, is worth $+2.91$ pp $[-13, +15]$, so the best-orientation rule's advantage over the cheap policy that reads no representation at all is $+4.56$ pp, nothing or less in $20\%$ of draws. Choosing the cut on the cells it is then scored on pays $+8.57$ pp: the $+1.10$ pp gap $[0, +13]$ is the price of a retrospectively chosen threshold, and it is what §9's "chosen on those same cells, so a lower bound and not a validated classifier" costs, measured. Per-cell hindsight is worth $+10.38$ pp $[+1, +19]$. Out of sample the rule freezes 16 of the 31 cells and puts 8 of them on the wrong side of its own decision, against 5 for the in-sample cut – the same count Figure 1b reports for the best threshold, reached here by maximising saving rather than by minimising misplacement, which are different objectives that happen to agree on this data.

**What we take from it, and what we do not.** We do not claim this refutes a CKA rule: a point estimate of $+7.47$ pp is not a refutation, and one policy family on 15 clusters cannot speak for a per-backbone cut, a multivariate rule or a continuous-shrinkage policy. What it supports is narrower and is the claim this paper makes: on these cells the evidence does not establish a saving stable enough to act on, and the direction in which acting would have paid is the opposite of the direction the mitigation literature points. Both readings survive the reviewer's objection to the null, because neither of them is the assertion that no relationship exists.

## REFERENCES

Table S16: **What would acting on CKA have saved?** The same 31 cells read as a decision rather than a correlation. A practitioner choosing per cell between full fine-tuning and a frozen encoder collects $\Delta_{\text{enc}}$ (positive = freezing better) on every cell they freeze, so *saving* = mean over all 31 cells of $\Delta_{\text{enc}} \cdot \mathbf{1}[\text{policy freezes}]$, in percentage points of MSE, against the default of always adapting. **The out-of-sample row is the only one a practitioner could have followed**: the threshold and its orientation are searched on the 14 training clusters and applied unchanged to the held-out cluster, so each cell's decision is made without seeing it. The row below it is the same search run on all cells at once, and the difference between them, $+1.10$ pp $[0, +13]$, prices the optimism in a retrospectively chosen cut. **The headline is the interval, not the point estimate**: the best-orientation rule's $+7.47$ pp comes with $[-4, +16]$ and is $\leq 0$ in 25% of draws, so it is a policy whose value these cells cannot establish, not one they refute. **And the orientation that earns it is the inverse of the prescribed one**: freezing where drift is *largest* – what forgetting mitigation actually recommends – is worth $-6.97$ pp $[-10, +14]$, and the row above it buys its point estimate by freezing the cells that moved *least*, a direction available only because we searched both. That rule still puts 8 of the 31 cells on the wrong side of its own decision out of sample (16 of them wanted freezing), against 5 for the in-sample cut. **This analysis is post-hoc and descriptive**: unlike Table 26, nothing here was pre-registered, and the family searched is a single global CKA threshold in either orientation – a per-backbone or multivariate rule could do better, and 15 clusters cannot rule that out. Candidate cuts are interior (midpoints between consecutive distinct CKA values), so the search cannot return *always freeze* or *always adapt* and have it counted as CKA informing the decision; both are priced as rows of their own. Intervals are 95% percentile cluster bootstrap (10,000/10,000 draws kept, a draw discarded when it leaves fewer than three distinct clusters), re-running the whole leave-one-cluster-out procedure inside each draw with duplicated clusters sharing one fold id. Every statistic is a mean of bounded per-cell quantities, so unlike $\Delta R^2$ in Table 26 these percentiles are reproducible – **to whole percentage points, which is why they are quoted that way**: at two decimals the bounds move across bootstrap seeds (this table's headline lower bound came out $-3.97$, $-3.61$, $-3.68$ and $-3.67$ on seeds 0, 1, 7 and 13), and `--seedcheck` checks every figure quoted here at its printed precision. The point estimates and the misplacement counts carry no seed at all.

| Policy | Saving (pp) | 95% cluster CI |
|---|---|---|
| Always adapt (fine-tune everything) – *the baseline* | 0.00 | *by construction* |
| Always freeze the encoder, ignoring CKA | $+2.91$ | $[-13, +15]$ |
| CKA rule *as prescribed*: freeze where drift is largest | $-6.97$ | $[-10, +14]$ |
| **CKA threshold, best orientation, chosen out of sample** | $+7.47$ | $[-4, +16]$ |
| the same rule, threshold chosen on the cells it is scored on | $+8.57$ | $[0, +18]$ |
| Oracle: freeze exactly the cells that wanted it | $+10.38$ | $[+1, +19]$ |